

MA 522: Partial Differential Equations

Course Code	MA 522
Course Name	Partial Differential Equations
L-T-P-C	3-1-0-4
Prerequisites	NA
Intended for	M.Sc./B. Tech./M.S./Ph.D.
Distribution	Core for M.Sc. in applied Mathematics and Elective for other discipline
Approval	10th Senate (Jun 2016)

Course Contents

- **Module I** [5 Lectures]
Introduction to PDE, First order PDEs, Solution methods for first order PDE.
- **Module II** [8 Lectures]
Classification of Partial Differential Equations, Cauchy Problem, Cauchy Kowalevski Theorem, Classification of Second Order Partial Differential Equations: normal forms and characteristics. Initial and Boundary Value Problems: Lagrange-Green's identity and uniqueness by energy methods.
- **Module III** [6 Lectures]
Methods of Solution, Methods of separation of variables, Characteristic method, Green's function, Fourier transform.
- **Module IV** [4 Lectures]
Stability theory, energy conservation and dispersion
- **Module V** [5 Lectures]
Laplace equation: mean value property, weak and strong maximum principle, Green's function, Poisson's formula, Dirichlet's principle, existence of solution using Perron's method (without proof).
- **Heat equation** [5 Lectures]
Initial value problem, fundamental solution, weak and strong maximum principle and uniqueness results.
- **Wave equation** [5 Lectures]
Uniqueness, D'Alembert's method, method of spherical means and Duhamel's principle
- **Module VIII** [4 Lectures]
Introduction to Hilbert Spaces of Functions, Sobolev spaces, Weak solution

Text Books

1. G. B. Folland, **Introduction to Partial Differential Equations**, Princeton University Press, 1995
2. L.C. Evans, **Partial Differential Equations**, Graduate Studies in Mathematics, Vol. 19, AMS, Providence, 1998.

References

1. . F. John, **Partial Differential Equations**, 3rd Edition, Narosa Publ. Co., 1979.
2. E. Zauderer, **Partial Differential Equations of Applied Mathematics**, 2nd Edition, John Wiley and Sons, 1989.
3. M. Renardy and R.C. Rogers, **An Introduction to Partial Differential Equations**, Texts in Appl. Math. 13, Springer, 1993
4. M.H. Protter and H. F. Weinberger, **Maximum Principles in Differential Equations**, Prentice Hall, 1967